

Rohan Varma

IS 310

Culture as Data Final Essay

Culture as Data: Understanding My Spotify Listening Habits

At the start of this project, my approach to the data was simple: I had listened to certain songs at certain times and had given them labels based on my emotional state. I thought I could just plot these patterns and understand them at face value. As I progressed, I realized that these categories were much more than simple classifications of music but rather something that I was creating myself. Instead of analyzing my listening behavior I became interested in what data creation for music could even look like.

The Manual Dataset: Data as Interpretation

I began this project by manually selecting about 75 songs from my listening history that I listen to on a regular basis. For each song I designated labels based on mood, energy level, and time of day. I also noted if I had skipped the song.

Although this process seemed straightforward at first, I found myself struggling to consistently define certain labels like mood. The same song could seem cheerful to me at one point and melancholic at another. This showed me that I wasn't just extracting data from my listening history but actively creating it through my interpretation. As discussed in literature about music and emotion, songs can express multiple emotional states simultaneously; in order to codify this within a single label, I was forced to simplify the actual emotion of the song in my head and lose a layer of nuance.

Scaling the Data Using the Spotify API

I then scaled the dataset to around 150 songs using the Spotify Web API by adding songs to my existing manual playlist. Scaling added a layer of computation to the project and made the process of generating the dataset much quicker and more manageable than my manual selections.

This transformation of my data collection into a more computational method introduced some trade-offs to the dataset. Whereas I spent significant amounts of time contemplating my own responses to each song in the manual stage, I had to move more quickly and rely on consistency in my interpretation to maintain coherence with the large amount of data I was generating.

What Changed as I Scaled the Data

Beyond merely adding data, the scaling process affected how I created the dataset. The labels I had created for the manual dataset stayed the same: mood, energy level, time of day, skip status, and source. But while my interpretation was somewhat loose in the manual stage, in the scaled version I had to establish very clear definitions for each category to apply them consistently across 150 songs.

This illustrated that scaling doesn't simply add to data, but it alters the production of data. The manual dataset yielded more specific but less transferable results, whereas the scaled dataset became more general but less detailed.

Computation also influenced how I created the data. In collecting the data through the Spotify Web API, the definition of many of my labels became reliant on how Spotify defines them. For example, the label 'popularity' is defined by Spotify and can vary drastically depending on when you retrieve this information. This again demonstrates that computation can complicate rather than clarify subjective experiences by layering different definitions on top of each other.

This particular revelation ties back to research on music and emotion such as Schaefer (2017), who emphasizes the multiple and combined emotions that music may elicit simultaneously; I was again forced to create single labels which stripped away from the complex realities of music.

Platform Influence: How Spotify Shapes Listening

Another surprising insight was the influence of Spotify itself as a platform. Initially, I assumed that the music I was listening to reflected purely my own subjective preferences; however, I soon realized that my dataset included music that I had added through playlists, received from Spotify recommendations, and encountered while listening to trending charts.

It occurred to me that my listening habits were not entirely of my own volition; they were shaped by the algorithmic and recommendatory design of the platform. The dataset thus became a reflection not only of myself but also of the role that the platform plays in my listening habits.

What the Dataset Reveals

Based on the dataset I developed, I am able to observe a few key trends in my listening habits. I am more likely to listen to either melancholic or more energetic songs at certain times of the day. I also found that I rarely skip songs that I personally curated on my own playlists. These tendencies seem to imply that my choice of music is indeed influenced by factors like the mood that I am hoping to create and the situation in which I am listening to it.

What the Dataset Conceals

The data, however, misses much of the complexity inherent to listening to music. Music is tied closely to memory and personal association which cannot be quantified within any meaningful framework. In my efforts to categorize songs by mood, I undoubtedly lost much of the depth and subjectivity in the listening experience by reducing the song to a set of labels and patterns.

Limitations of the Dataset

This dataset is limited in that many of the categories I used were subjective and potentially fluid. It is also only a snapshot of my own habits and can't be generalized to represent the typical listening habits of any user. Furthermore, the influence of Spotify's algorithm is present but cannot be explicitly seen within the data. Overall, it serves as an illustration that all cultural data will always have partial truths.

Ethical Considerations

Because this dataset was formed from personal music habits, no identifying information was included. My goal was to capture the music choices and related tags but to do so without revealing details of my identity or personal life. This is also of course related to the larger ethical considerations of how personal data is gathered and exploited on digital platforms.

Conclusion

My project changed the way I perceive data. I began the project with the perspective that data represents objectivity and measurability but now view it as being fundamentally bound by interpretation, computation, and the specific system in which it was produced. Data can represent patterns within the context of cultural experience, but in doing so, often loses the very complexity and nuance that are central to culture itself.

References

Schaefer, H. E. (2017). Music-evoked emotions: Current studies. *Frontiers in Psychology*, 8, 2239.[LINK](#)

Spotify. (2026). *Spotify Web API*. <https://developer.spotify.com/documentation/web-api/>

Spotify. (2026). *Personal playlist dataset used in this project* [Data set]. Spotify.

Eerola, T., & Vuoskoski, J. K. (2011). A review of music and emotion studies: Approaches, emotion models, and stimuli. *Music Perception*, 30(3), 307–340.[LINK](#)